

Steps to Creating Web API using .NET with Visual Studio

1. Open Visual Studio
2. Create a new project → Select ASP.NET Core Web API Project Type/Template → Go to Next
3. Give the Project Name & Select the location where you want to store or save the project → Go to Next
4. Select the Latest .NET Framework & all things must be remain as it is → Go to Create

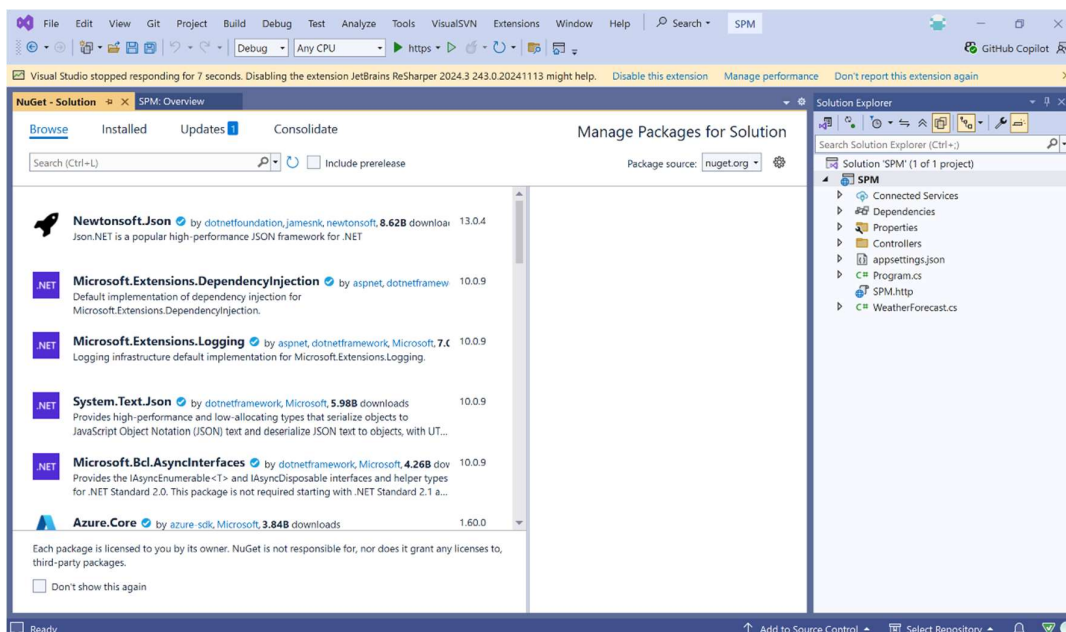
Project Structure

ProjectName

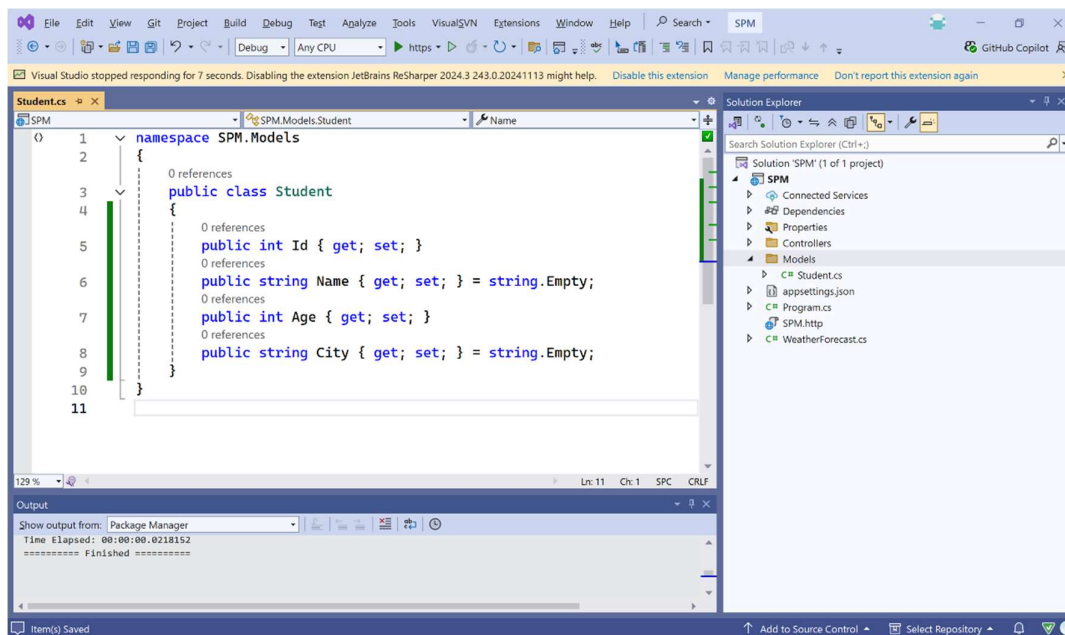
- Controllers
 - StudentController.cs
- Data
 - AppDbContext.cs
- Models
 - Student.cs
- Program.cs
- Appsettings.json

Next Step - Install Packages

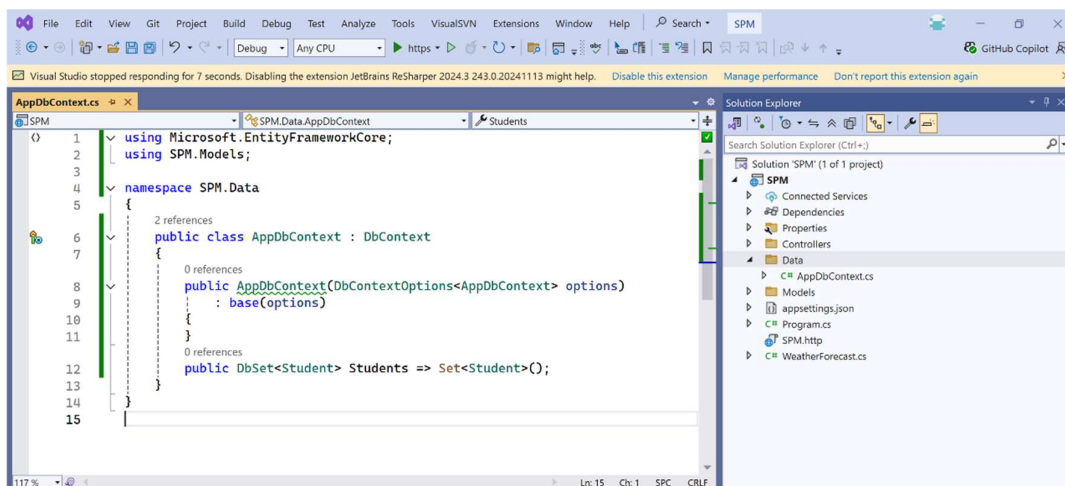
1. Go to Menu Tools → NuGet Package Manager → Manage Nuget Packages for Solution



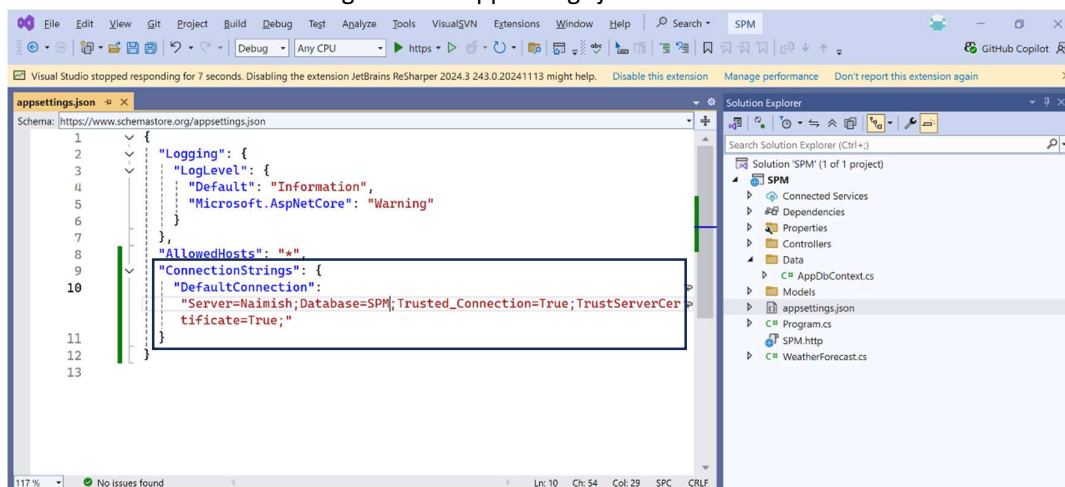
2. Install below mentioned packages in your project
 - a. Microsoft.EntityFrameworkCore.SqlServer
 - b. Microsoft.EntityFrameworkCore.Tools
 - c. Scalar.AspNetCore
3. After installing packages in your project check/verify those in your project directory, Go to solution explorer → ProjectName → Dependencies → Packages
4. Now Start with Models (Database Tables), to create models in your project first create a folder and give name to that folder to "Models".
5. Add One Class file in it having name **Student.cs** (Right click on Models folder → Add → Class)



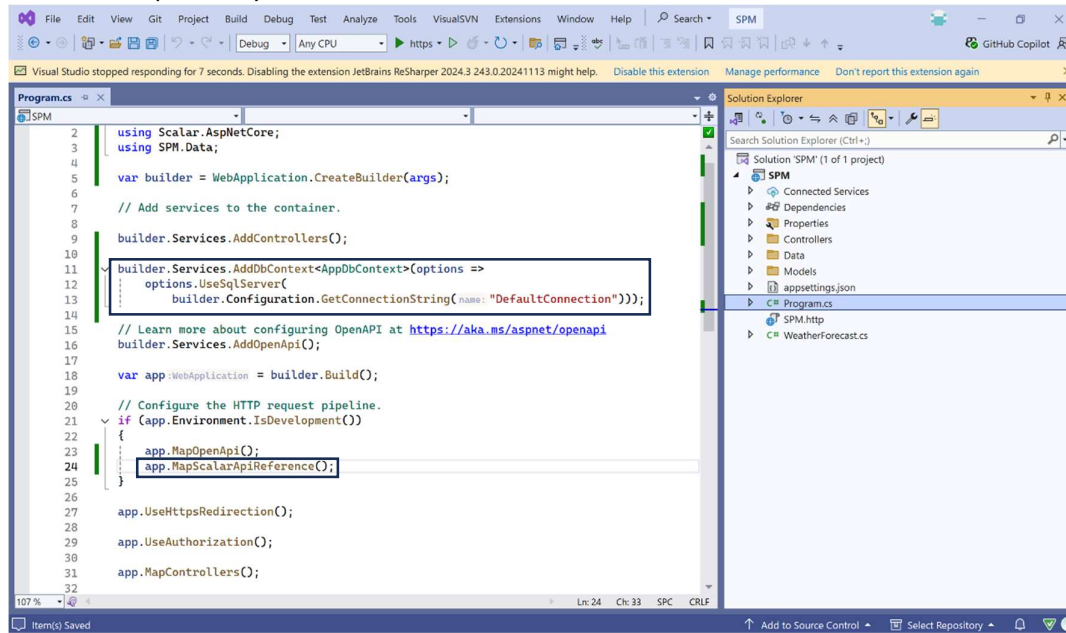
6. Now Start with AppDbContext.cs [Responsible for your project related registers means in your project many tables are there so all that things managed by this class].
7. Create **Data** Folder → Add One Class file in it having name **AppDbContext.cs** (Right click on Data folder → Add → Class)



8. Connection string (We need to give connection string because which database?, which server?)
9. Do add below mentioned things in to an appsettings.json file.

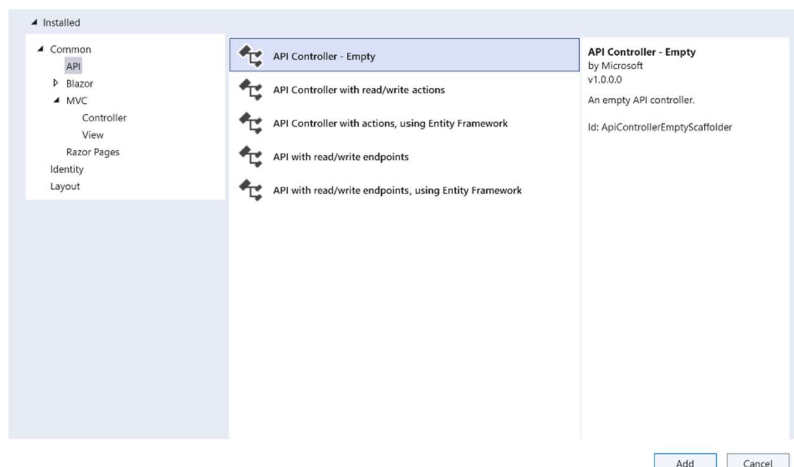


10. Now our main file program.cs, we have to register the connection string & scalar (API testing) reference/dependency to it.

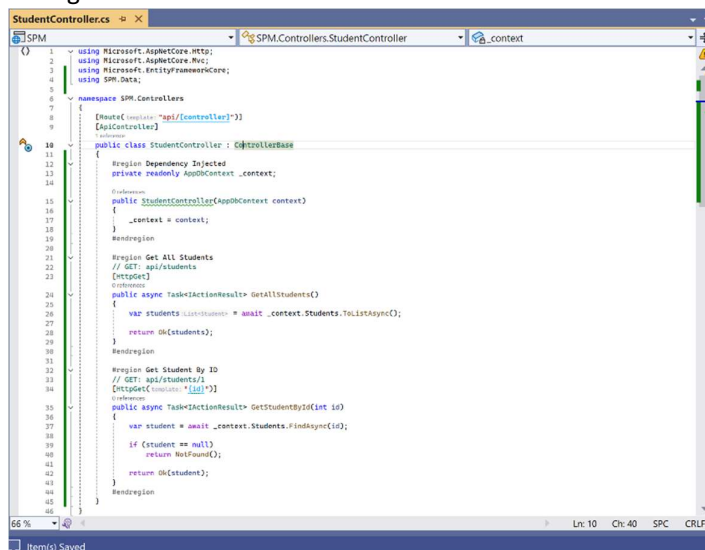


11. Now inside Controller folder, add one Controller for Student CRUD. (Right click on Controller folder → Add → Controller → Select API Controller – Empty → Give Name to **StudentController.cs**)

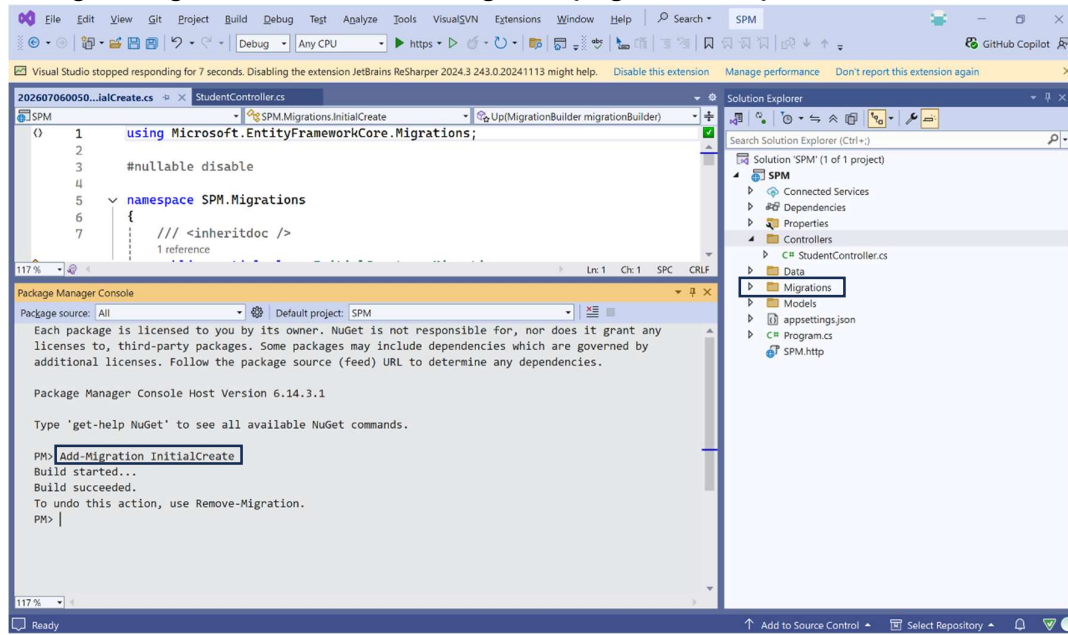
Add New Scaffolded Item



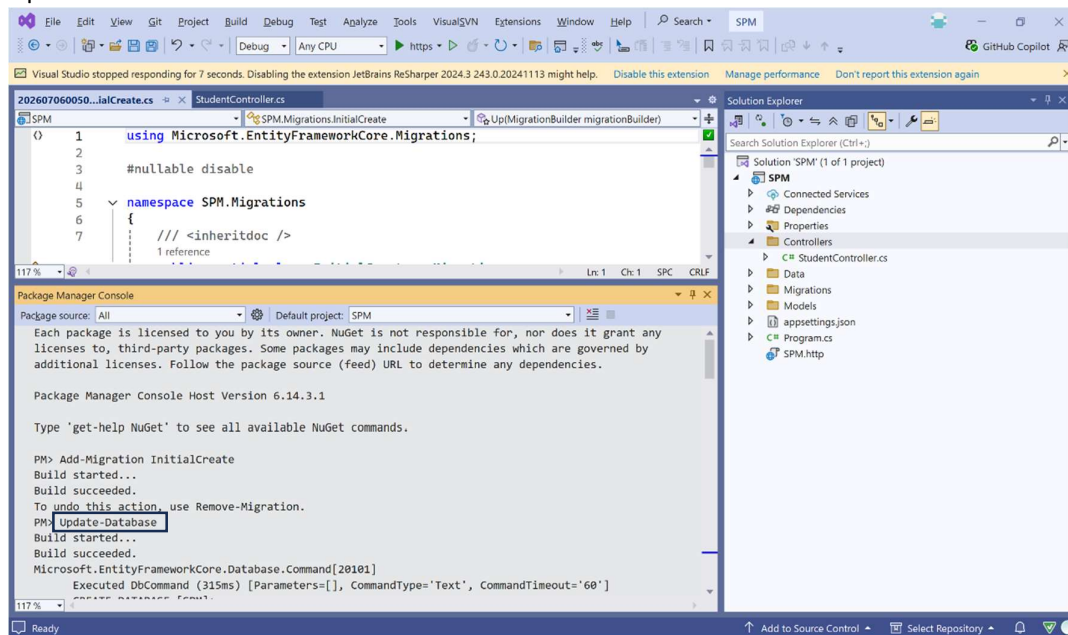
12. Now go the StudentController.cs file.



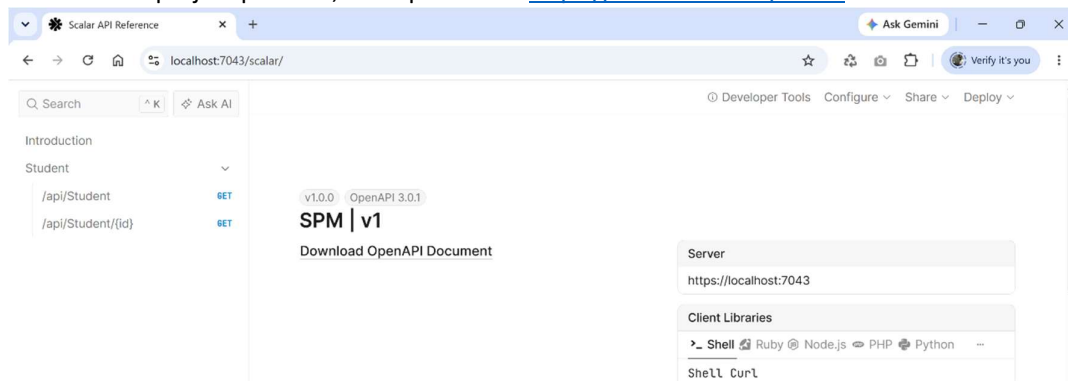
13. Now Create Migrations (Auto Database & Tables creation), Go to Tools Menu → NuGet Package Manager → Package Manager Console & write **Add-Migration {MigrationName}**



14. Now final step these all things must be sync with our database so in last we have to give command as **Update-Database**



15. Now run the project press F5, and open scalar <https://localhost:xxxx/scalar>



16. See the result

The screenshot displays an API client interface with the following details:

- URL:** `https://localhost:7043 /api/Student`
- Method:** `GET`
- Status:** `3.7s 200 OK`
- Response Headers:** 3 headers are listed.
- Response Body:** A JSON array of two student objects.

```
1  [
2    {
3      "id": 1,
4      "name": "Rahul",
5      "age": 21,
6      "city": "Rajkot"
7    },
8    {
9      "id": 2,
10     "name": "Priya",
11     "age": 20,
12     "city": "Ahmedabad"
13   },
14 ]
```